Student Name Student ID No.: Section		Computer Engineering Department Microprocessors Lab – 30 points Mid Exam

2) Write a program in assembly language to find the total of memory locations that ar greater than 10, storing the result into SUM, and push them into the stack.

```
DB 2,5,8,12,17,1,0,4,13
```

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```
INC CX
INC CX
CMP CX,8
JB  L
```

5.
```
MOV AX,27
MOV BL,4
MOV CL,3
DIV BL
XOR AH,FFH
```

AL= ; AH=

6. Find the content of register AL after execution of the following code:

```
MOV AL,0FH
MOV CL,2
SHR AL,CL
ADC AL,1
```

AL=

A

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A) SI
B) CS
C) SS
D) SP

7. The underlined number represents

AL

8 f

8. MOV SI,7
MOV AL,0
L2: CMP X[SI],10

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Multiple Choice Questions: **(15) points**

1	2	3	4	5	6	7	8	9	10

1. What is the type of addressing modes used in instruction MOV CL,[BP][SI]

A. Indirect B- base relative C- Base Indexed D-Direct Indexed

2. Suppose X defined as below; what is the content of X after execution of the following instructions:

```
X  DD 335647FBH,22H
MOV BX,OFFSET X
MOV AL,34H
ADD [BX+1],AL
```

X=

3. Find the content of AL and zero flag (ZF) and carry flag (CF) after execution of the following code:

```
MOV AL,FCH
ADD AL,3
ADC AL,3
```

AL= ; ZF

CF=

Find the value of AX after execution the following assembler codes

```
 MOV  CX,1
 MOV  AX,0
L: ADD AX,CX
 INC  CX
 INC CX
 CMP  CX,8
JB   L
```

AX=

```
AX,27
L,4
L,3
```

AL= ; AH=

content of register AL after execution of the code:

AL=

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```
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
```

```
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
[illegible]
```

```
B
ORG 1000H
MOV AX, 1036H
MOV BX, 20H
PUSH AX
PUSH BX
XCHG AH, BL
ADD AX, BX
POP AX
POP BX
HLT
```

30. The result of multiplication instruction MUL BX is stored in register pair:

A. DX,AX

C. CX,AX

Question#2(15 points)

Given memory locations X defined as below

X DB 51,15,76,40,90,45,13,20

to an assembler codes find the memory locations that are divided by 3;

- Find the sum of them; storing the result into SUM

- Replace them by ° character

push them into the stack

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